

OdeD at GroundLM 2026 Shared Tasks: Reading the Scorer for Literature-Grounded QA

Omar Saqr

The American University in Cairo
omar_saqr@aucegypt.edu

Mostafa Gaafar

The American University in Cairo
mostafa21314@aucegypt.edu

Abstract

We describe team **OdeD**'s submission to LitTraceQA, the literature-grounded question answering task at GroundLM 2026. The task scores three coupled outputs: paper identifiers, coarse evidence locators, and answers in multiple-choice or table form. Our best official score on the 71-question held-out test split is **0.7649**, against **0.4563** for our first submission and **0.5519** for our best fully automated one. Little of that improvement came from the retrieval-and-reading pipeline we began with; most of it came from two other sources. Reading the released evaluator showed the metric to be more asymmetric than it appears: paper F1 and evidence F1 carry weight 1/3 each while each of the three answer metrics carries 1/9. The evidence key is also a coarse tuple, so a correct value on a correct page scores zero when the *source type* or the visible object id is wrong. The 55-example development split then served to recover the dataset's *annotation conventions* rather than as a tuning set, and our own submission history became a measuring instrument. Each macro metric is a mean of a small number of rational per-question values, so a controlled change to one prediction file often admits a single arithmetic explanation. Two predicted deltas returned exact to four decimals. We report the conventions this recovered, six heuristics it refuted, four bugs it exposed in our own verifiers, and the limitation that much of our final score reflects per-question auditing and leaderboard feedback rather than a system that would generalise.

1 Introduction

LitTraceQA (Liu et al., 2026) asks a system to answer a research question by returning three things at once: the canonical identifiers of the paper or papers involved, coarse evidence locations inside those papers, and the answer in a requested format. Because these are scored separately, a system can be visibly right for the wrong reasons: a correct

multiple-choice answer attached to the wrong paper, or the right paper with a locator pointing at the wrong page. It is one of the two GroundLM 2026 shared tasks, alongside GoldenViewVQA (Wang et al., 2026a), which asks the same provenance question of multi-view driving scenes; we entered LitTraceQA only. Wang et al. (2026b) report the tasks and the participating systems.

Our participation had an unusual shape. We built a conventional pipeline: dense retrieval over the released metadata pool, cross-encoder reranking, an LLM selector, and a vision-capable reader for tables and figures. It reached an official 0.4563, improved to 0.5519 with selection and visual reading, and then stopped. Every subsequent gain, to 0.7649, came from work better described as *instrument reading* than as modelling: studying the scorer, recovering the annotators' conventions from the development split, and using controlled submissions to identify which individual predictions were wrong. The findings seem more transferable than our architecture, so the paper is organised around them.

Contributions.

1. A decomposition of the official metric showing that two-thirds of the score is *provenance* rather than answering, and that the set-based evidence and table metrics make the cost of hedging exactly computable (§2).
2. Eight annotation conventions recovered from 55 development examples, each reported with the count it rests on, and the two we acted on directly in test submissions (§4).
3. *Score-guided attribution*: using a macro metric over a small test split to prove an individual prediction right or wrong, with two predicted deltas confirmed exactly (§5).
4. Six negative results and four bugs in our own

instrumentation, reported so that others do not repeat them (§8).

5. An error taxonomy for grounded scientific QA in which the dominant failure is *the right paper and the wrong row* rather than hallucination (§7).

2 The Metric Is Asymmetric

The released evaluator computes

$$\text{answer} = \frac{1}{3} (\text{mc} + \text{row_f1} + \text{cell_acc})$$

$$\text{overall} = \frac{1}{3} (\text{paper_f1} + \text{evidence_f1} + \text{answer})$$

so paper F1 and evidence F1 carry weight 1/3 each, while multiple-choice accuracy, table row F1 and table cell accuracy carry 1/9 each. Treating answer accuracy as the objective therefore puts effort in the wrong place. On our first submission multiple-choice accuracy was already 0.68 while evidence F1 was 0.3587. Figure 1 gives the system and where each stage moved the score; Figure 2 shows how the components moved across our scored submissions, and Table 1 gives one question end to end.

Evidence keys are coarse. Predictions and gold are compared as sets of tuples $(\text{paper_id}, \text{source_type}, \text{page}, \text{object_id})$, where object_id is a normalised visible identifier for table, figure, equation_algorithm and citation_context and is *empty* for text_span. Three consequences shaped our work. (i) A `text_span` matches on paper and page alone, making it the cheapest correct guess when the type is uncertain. (ii) Citing the right page with the wrong type scores zero: “we found the answer” is not “we located it as the annotator did”. (iii) Emitting N predictions against G gold items with C correct gives $F_1 = 2C/(G + N)$, so the cost of hedging is arithmetic rather than taste. We use (iii) throughout.

Cell accuracy ignores row precision. Row F1 is a set F1 over row-key tuples, so duplicate row keys silently collapse in the evaluator’s dictionary. Cell accuracy iterates over *gold* rows only: cells in a predicted row whose key does not match gold are never examined, and unmatched gold rows count their cells as wrong. Cell accuracy is therefore bounded by row *recall* and indifferent to row precision: an extra candidate row can cost row F1 but

Q “...what MATH score does the BlueLM-3B-based MLLM reach when fine-tuned on Cambrian 7M plus an extra 2M pure-text samples...?”

gold_papers: iccv2025_01015
evidence: (iccv2025_01015, table, p3, "Table 3")
answer.multiple_choice.gold: B

Page 3 carries both Table 2 (“expand the Cambrian-7M dataset with 2M pure text data”) and Table 3 (“using Cambrian 2.5M”). The question’s wording matches Table 2 almost verbatim, and gold cites **Table 3**. Choosing Table 2 cost a full 1.0 of evidence F1 on this one question, because `object_id` is part of the key.

Table 1: One question, the three coupled outputs, and why the evidence metric is unforgiving: the answer and the page are right, and the item still scores zero if the visible object id is wrong.

can never cost cell accuracy. Including this term moves the break-even hit rate for adding a candidate row from about 60% to **26%**, which changed several of our decisions.

3 System

Retrieval and selection. The pool contains 27,487 papers with title, abstract, venue and PDF URL. We embed the question and title–abstract text, retrieve a candidate list, rerank with a cross-encoder, and pass the top candidates as a numbered list to an LLM selector that returns indices. Reranking moved paper F1 from 0.410 to 0.490 on a development family; the LLM selector with a cap of three papers per question was our best configuration.

Shortlist recall, measured separately to locate the bottleneck, is not the constraint: on a test-like development family, recall of the gold paper was 0.846 at rank 1 and 0.962 at rank 20, and *flat* out to rank 200. Widening the shortlist was therefore pointless and we did not pursue it.

Reading. Selected PDFs are read locally with PyMuPDF for page text, caption indices and page rasterisation. Answer generation is a per-question prompt carrying the question, the requested answer types, the supplied table schema, and page text or rendered page images. Table questions use a two-stage scheme in which row keys are extracted from the question before values are read, guarded by a check that every extracted key occurs in the question; without that guard the stage fabricated entities and cost 0.129 overall.

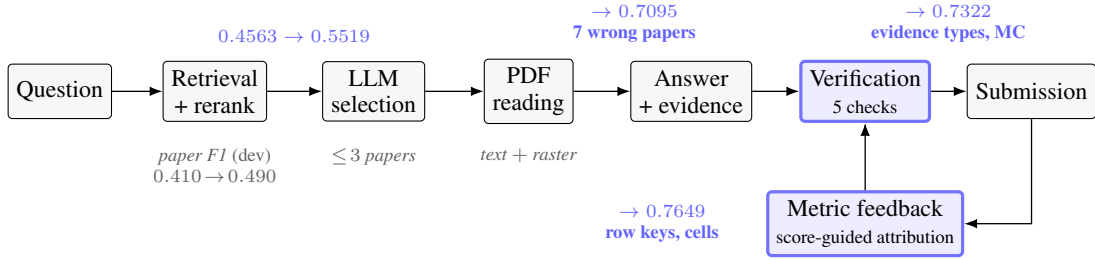


Figure 1: Our system and where each stage moved the official score. The unshaded left-to-right path is the pipeline, which plateaued at 0.5519. The two shaded boxes produced the rest: a verification pass that checks every prediction against the source PDF, and a feedback loop in which the evaluator’s macro metrics are decoded arithmetically to identify *which* individual predictions are wrong (§5). Every score is an official test score except the retrieval annotation marked *dev*, a development-set ablation of the reranker.

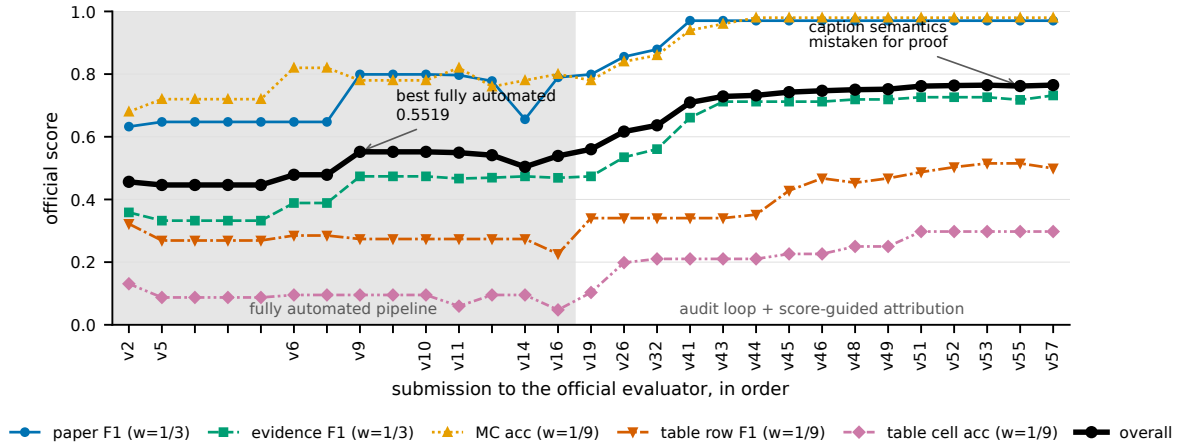


Figure 2: Official evaluator score for the 29 scored submissions we hold evaluator records for, in order. Ticks are labelled on a run’s first appearance; unlabelled ticks are re-uploads of an identical file, and one run has no surviving local prediction file. The shaded region is the fully automated pipeline; from v19 onwards every submission involves per-question intervention. The two components weighted 1/3 account for almost all of the movement, while the two table components weighted 1/9 stall between 0.05 and 0.52 and hold most of our remaining headroom. v55 regressed: we had labelled a caption-semantics inference “confirmed” when it was a heuristic, and reverting it returned exactly the +1.0 of evidence sum the arithmetic predicted. Drawn from `results/official_scores.csv`, which is built from the evaluator’s submission log.

Audit loop. The component responsible for most of our score is a procedure rather than a model. Each candidate submission is checked question by question against the source PDFs: does the cited page contain the claim, does the cited object id appear in that page’s captions, does every emitted paper mention an entity the question names, and does the chosen option match the row and column the question actually asks about. Adjudication was performed by an LLM agent reading page text and rasterised pages, with every decision recorded against a specific page of a specific PDF. This is a human-in-the-loop system in the sense that matters for interpreting our score, and we return to it in §10.

4 Recovered Annotation Conventions

The development split has 55 examples with gold papers, evidence and answers. As a tuning set it is too small to separate configurations. As a corpus describing *how the annotators behaved* it was the most valuable resource in the task. Table 2 collects the conventions with their denominators, because several rest on small samples and a reader should see that rather than take them on trust.

Evidence keys match the gold paper count.

Distinct evidence keys versus gold papers gives (1, 1) 24 times, (4, 4) 19 times, plus (3, 3) and (9, 9), so the counts agree on 45 of 55 questions. On 37 the mapping is strictly one key per paper; the eight remaining are four-paper questions where

Convention (counted on 55 dev questions)	support
Distinct evidence keys = gold paper count	45/55
Gold table rows = gold paper count (multi-paper)	8/8
Question names a figure $\Rightarrow$ gold has figure	10/10
Question names a table $\Rightarrow$ gold has table	3/3
Asks an equation/loss $\Rightarrow$ gold has equation_algorithm	4/7
Cited page carries a table caption $\Rightarrow$ gold type is table	52/68
Gold pairs a text_span with a same-page table	1/64
Gold page is the <i>earliest</i> occurrence of the value	17/84

Table 2: Conventions recovered from the development split, recomputed for this version by `results/count_conventions.py`. Row 1 counts questions whose number of distinct evidence keys equals their number of gold papers; the stricter reading, one key per paper with no paper left uncited, holds for 37/55. Row 8 counts gold items whose value occurs on more than one page of the cited paper. Rows 2 and 3 were the conventions we acted on directly in test submissions (§4). The last two rows are why we did *not* ship a positional page-rewrite rule or a same-page text_span pad.

one paper carries two keys and one is cited by none. Gold evidence is provenance *per paper*, not an exhaustive list of supporting locations, and we sized our evidence sets accordingly.

Source type follows the question’s wording. Conditioning gold’s source type on the question’s vocabulary is predictive on the development split, though on small denominators (Table 2). Acting on it produced our largest evidence gain: we had been emitting *no* equation_algorithm items at all, and adding them alongside text_span on six equation-form questions coincided with official evidence F1 moving from 0.6610 to 0.7121 (that submission also re-aimed two pages, so the delta is not attributable to the type change alone). Where gold’s type is genuinely mixed, carrying both is the right arithmetic: against a single gold item, two predictions with one hit score 2/3 where committing to one scores 0.4 in expectation at $P(\text{equation}) = 0.6$.

Gold table rows equal gold paper count. Every multi-paper table question in the development split has exactly one gold row per paper (8 of 8, no exceptions). This does more than cap row F1: if

gold has two rows, then gold’s two keys *cannot* be two of our four quantity descriptors, because a two-row table cannot key its rows by four separate quantities. Three of our test predictions violated the constraint and were, by that argument, scoring zero.

Row keys reproduce printed labels. Where a question names entities, gold keys are those names lowercased, keeping qualifying parentheticals. Where it implies variants of one paper, gold uses the labels *printed in that paper’s own table*. One development question keys its rows w/ ground truth and w/ cube rcnn. Checking every test row key against its source this way found four mismatches. The largest was a retrieval question whose gold keys carry the paper’s +Fake2 suffix rather than the bare dataset names we had used. Correcting it took that question from row F1 0 to 1.0 and cell accuracy 0 to 1.0 in a single step. We *added* the variants rather than substituting them, which is what guaranteed the eight cells behind those rows would be scored under either convention; a wrong bet would have zeroed them too.

Evidence pages are not the first mention. Gold’s page histogram peaks at pages 6–7 (65 of 149 items) with only 5 of 149 on page 1, while our predictions put 40% of their items on pages 1–3. The obvious correction is nevertheless wrong: among the 84 gold items whose value appears on more than one page of the cited paper, gold is the earliest occurrence 17 times, somewhere in the middle 61 times, and the latest 6 times. There is no positional rule to learn, only a semantic one. The statistic served to triage items for manual review; it was never shipped as a rewrite.

5 Score-Guided Attribution

The task permits five evaluator submissions per task per day. Because each macro metric is a mean of a small number of rational per-question values, a submission that changes one thing often has exactly one arithmetic explanation, so the feedback identifies *which* predictions are wrong rather than merely whether the file improved.

Worked example. Table row F1 sat at exactly 0.340476 for five consecutive submissions. Changing two questions’ row keys then moved the macro by +0.011111, or +0.2333 over the 21-question

sum. That shift decomposes uniquely: one question went from zero matched rows to one while its row count went from 2 to 4 ($2 \cdot 1/6 - 0 = +0.3333$), and a second question’s added candidate missed ($2 \cdot 1/5 - 2 \cdot 1/4 = -0.1$). This established that gold’s paper row keys are short names rather than full metadata titles, and that one specific candidate string was wrong.

Reading exact fractions. `table_cell_accuracy_micro` is `cell_correct/cell_total` and `cell_total` is a property of gold. Read as an exact fraction it gives 23/87 and later 31/87: gold has exactly 87 comparable cells across the 21 table questions, and the +8 was one question’s eight cells.

Predictive accuracy. Two predictions made before submitting came back exact. Dropping rows proven not to match was predicted at +0.2952 over the row sum and returned +0.2952; restoring three row sets was predicted at +0.2500 and returned +0.2500. A third prediction was wrong by -0.25 . Three of the changed questions had identical key sets before and after, so their contributions had to cancel; the shortfall therefore forced the conclusion that a change we had reasoned was “free” had discarded a matching key. We reverted it and recovered the exact amount.

Zero-cost slots. The same arithmetic identifies predictions that cost nothing to change. If a question has $C = 0$ its row F1 is zero for any N , so replacing its keys is free. Row F1 was unchanged to six decimals after we added one row to each of two questions; adding a row can only leave F1 fixed when $C = 0$, and no other combination of their possible deltas sums to exactly zero. Both therefore score zero on rows, meaning FVT and AP-Attack do not match gold *despite being exactly what the question calls those things*.

Disclosure. This is adaptation to the held-out split. It is within the stated submission rules, and we report it because it materially explains our final number; scores obtained this way should not be read as an estimate of how the underlying system would perform on unseen questions.

6 Official Results

Table 3 reports official evaluator output for our scored submissions on `littraceqa-test` (71 questions). Every figure is as returned by the eval-

uator; no number in this paper is computed by us from private labels. We did not submit to the optional `littraceqa-test-extra` diagnostic track.

Where the score came from. Between our first and best submission, paper F1 rose by 0.338 and evidence F1 by 0.373; multiple-choice accuracy rose by 0.30 but contributes at one third the weight. Table row F1 and cell accuracy remain our weakest components at 0.4992 and 0.2976 and together hold the majority of our remaining headroom.

Automated verifiers. Five checks ran over every candidate file. (1) An *object-id verifier* regexes captions from each page and confirms the cited `table_id/figure_id` exists there; 48 of 49 of our ids passed and the single failure was a real error. (2) A *name-presence check*: does each emitted paper’s text ever mention an entity the question names. A paper that never writes “ERASE” is not the ERASE paper. (3) The same check against all 27,487 titles, which finds papers never retrieved and therefore unreachable by any text-based check. (4) A *crowded-page check* flagging table items on pages carrying more than one same-type caption, where “the id exists on this page” does not imply the right table. (5) A *dead-run guard* failing any submission with all-null table rows; this caught two files silently produced after our API quota was exhausted mid-run, with no error anywhere in the trace.

7 Error Analysis

The dominant failure is the right paper, wrong row. Of the answer errors we found and corrected, almost none are hallucinations. They are values genuinely printed in the cited paper, one row or one column from the one asked about: the unfinetuned base-model row instead of the fine-tuned one; RPEt where RPEr was asked; the MC1 column where the question says MC3; inter-annotator agreement where the question says intra-annotator; a *Pearson* column read as a Kendall one; the *None* baseline row’s accuracy reported as the proposed method’s macro-F1; a *cross-check-enabled* row where the question says disabled. This is why three separate value-presence triages we built returned mostly false positives — every candidate value *is* in the paper — and why reading row and column headers is the only check that works.

run	Submission	paper F1	evid. F1	MC	row F1	cell acc. (macro)	overall
v2	initial pipeline	0.6324	0.3587	0.68	0.3221	0.1310	0.4563
v6	+ selection set size, hosted reader	0.6474	0.3887	0.82	0.2849	0.0952	0.4787
v9	+ LLM selector, visual table reading	0.7991	0.4737	0.78	0.2738	0.0952	0.5519
v19	+ hand-authored table row keys	0.7991	0.4737	0.78	0.3405	0.1032	0.5602
v26	+ answer audit against the PDFs	0.8554	0.5347	0.84	0.3405	0.1984	0.6166
v32	+ audit continued	0.8789	0.5606	0.86	0.3405	0.2103	0.6366
v41	+ seven paper-set corrections	0.9704	0.6610	0.94	0.3405	0.2103	0.7095
v43	+ equation_algorithm evidence	0.9704	0.7121	0.96	0.3405	0.2103	0.7287
v44	+ figure-counting answers	0.9704	0.7121	0.98	0.3516	0.2103	0.7322
v45	+ short-name table row keys	0.9704	0.7121	0.98	0.4283	0.2262	0.7425
v46	– rows proven not to match	0.9704	0.7121	0.98	0.4675	0.2262	0.7468
v48	+ batched cell rewrites, 3 candidate rows	0.9704	0.7192	0.98	0.4534	0.2500	0.7502
v49	– the 3 candidate rows, all missed	0.9704	0.7192	0.98	0.4675	0.2500	0.7518
v51	+ printed-label row keys	0.9704	0.7262	0.98	0.4873	0.2976	0.7617
v52	– rows proven not to match (2)	0.9704	0.7262	0.98	0.5032	0.2976	0.7634
v53	+ three row sets restored	0.9704	0.7262	0.98	0.5151	0.2976	0.7647
v55	+ caption-semantics table swap	0.9704	0.7178	0.98	0.5151	0.2976	0.7619
v57	– swap reverted, + two key unions	0.9704	0.7319	0.98	0.4992	0.2976	0.7649

Table 3: Official evaluator results on `littracqa-test`, in submission order; each line names the change from the line above. Listed are the submissions marking each distinct stage of our progression, plus v55, which regressed and which we report rather than only our best. Our submission history is longer than the table: the evaluator’s submission log records *at least* 29 scored runs, several of them re-uploads of an identical file. The log as we were able to export it is in the repository, and it is the source for every number in this table. v9 is our best **fully automated** submission; every later row involves per-question intervention. Paper F1 and evidence F1 carry weight 1/3 each in *overall*; MC, row F1 and cell accuracy carry 1/9 each. Cell accuracy is the evaluator’s *macro* value; the micro value it also reports differs (for v57, $0.3563 = 31/87$). Components are rounded to four decimals, so recomputing *overall* from them can differ in the last digit.

Wrong papers hide behind plausible titles.

Seven paper sets were wrong and every one was found by the presence checks rather than by re-running selection. Two required fetching a PDF that had never been retrieved, so no text-based check could have reached them; one was a question answered “all three papers” while citing a paper that was none of the three the options named.

Figures need rasterising. Four questions ask for counts existing only in a figure. Rendering those pages at $2\times$ and reading them settled all four and corrected one answer: a paper states in prose that it classifies models into *eight* categories, and its pie-chart labels agree, where we had answered six.

Annotator-invented strings are not recoverable.

Where a table’s row-key column is a descriptor the annotator invented (`quantity`, `setting`, `attribute`, `metric`), we never matched it. Our record on guessing such strings is **0 for 13**: three candidate row keys in one submission (all missed), four successive forms of a single cell (all missed), two structural conventions on three questions (both missed), two convention unions (both missed), and the literal words the question uses for two entities (FVT, AP-Attack; both score zero). We report this as a limit of the approach rather than an un-

explored lead, and it is the main reason our table components stalled.

8 Negative Results

We list heuristics that seemed reasonable, were measured, and were dropped.

- **Padding paper sets.** From one submission’s precision and recall alone, the gold set sizes on test are derivable as 1 or 2 (3 and 4 are arithmetically impossible), and our selector is size-calibrated: an uncalibrated selector cannot reach the observed precision even if its second pick were always gold. Padding singletons was refused.
- **Full-text indexing.** Content-defined multi-paper questions look like they need body text, but only 4% of the relevant development family is answerable from body text alone, and the shortlist was already saturated.
- **Re-aiming evidence at later pages.** Motivated by the page histogram; measured at $+0.013$ evidence F1 on development, below our shipping bar, and explained by the middle-position statistic.

- **Carrying both row-key spellings everywhere.** Measured at -0.020 row F1 on development, because there the bare keys are usually already right and the alternative never hit. We restricted hedging to keys we could *prove* wrong.
- **Converting text_span to table on caption-bearing pages.** The conditional is real (52/68), but all seven of our text_span items on such pages are verified prose, so there was nothing to convert.
- **Caption semantics as proof.** The one submission on our best-score path that scored below the one before it ($0.7647 \rightarrow 0.7619$). A question asks for a setting whose wording matches Table 2’s caption almost verbatim (“expand the Cambrian-7M dataset with 2M pure text data”) while Table 3 reads “using Cambrian 2.5M”; gold cites **Table 3**. Reverting returned exactly $+1.0$ of evidence sum, confirming it. Caption semantics is a useful heuristic and we had labelled it “confirmed”, which was an error of epistemics rather than of arithmetic.

Four bugs in our own verifiers. Each briefly produced a conclusion we believed. (i) A multiple-choice triage stripped whitespace before matching numbers, which glues adjacent table cells together, so `16,825,000 5,800,000` became one token and both halves were reported absent; every “value not in this paper” verdict was worthless until whitespace was collapsed rather than removed. (ii) A caption regex matched in-text references such as “...in Table 3.”; requiring the caption at line start raised the measured type conditional from 71.6% to 76.5%. (iii) A prediction-file diff compared a key that prediction files do not carry, so both sides read `None` and it reported “0 of 71 identical” when the true answer was 26. (iv) We reported a PDF as unreadable and concluded the pipeline never saw it; probing all 112 selected papers in isolated subprocesses gave *zero* unreadable, and the crash was resource exhaustion in our own script. An analysis pipeline is itself an experimental instrument, and ours was wrong four times in ways that stayed invisible until checked.

9 Setup and Disclosed Resources

Data. The released LitTraceQA metadata pool (27,487 papers), the development split (55) and the input-only test split (71), from the official Hugging

Face dataset under CC BY-NC 4.0 (Liu et al., 2026; LitTraceQA shared task organisers, 2026). PDFs were fetched from the publisher URLs in the released metadata. No additional annotated data was used and no synthetic training data was generated. All held-out scores come from the official evaluator Space (GroundLM 2026 shared task organisers, 2026); we never held the test labels.

Models and APIs. Selection and reading ran on the Gemini free tier (Google, 2026), rotating over `gemini-flash-lite-latest`, `gemini-3.5-flash` and `gemini-flash-latest` in that order, with `gemini-3.7-flash` used only in one selection experiment. Rotation is forced by the per-model daily quota: `gemini-flash-latest` resolved to `gemini-3.6-flash` with a measured 20 requests/day, and a 71-question run needs roughly 210 calls. Decoding was greedy (temperature 0). Two names are floating aliases, so the exact weights behind them cannot be pinned: our runs fall between 16 and 19 August 2026, and we did not record the resolved version returned per call.

Text-only selection experiments used a local `Qwen3.6-27B-UD-Q4_K_XL GGUF` served by `llama.cpp`’s `llama-server` (`--n-gpu-layers 42 --ctx-size 12288 --parallel 1 --cont-batching --jinja --cache-type-k q8_0 --cache-type-v q8_0`). Decoding used temperature 0.7, `top_p` 0.9 and three samples per question, with output grammar-constrained to our read schema and Qwen3 thinking switched off. `Qwen3-8B` (Qwen Team, Alibaba Group, 2025) was the `transformers-path` default. We did not record the `llama.cpp` build. Retrieval used BAAI/`bge-large-en-v1.5` for dense embeddings (Beijing Academy of Artificial Intelligence, 2026a) and BAAI/`bge-reranker-base` as cross-encoder reranker (Beijing Academy of Artificial Intelligence, 2026b). The audit loop of §3 was carried out by an LLM agent (Anthropic Claude (Anthropic, 2026)) reading page text and rasterised pages; we did not record the dated model version. We fine-tuned and trained nothing, so there are no checkpoints of ours to release.

Software and compute. Python, with PyMuPDF (Artifex Software, Inc., 2026) for page text, caption indices and rasterisation. Local checks used the organisers’ `evaluate.py` and their

`validate_submission.py` extended with the dead-run guard of §6. All work ran on one workstation; the only GPU use was local inference and reranking.

Reproducibility. The repository holds the pipeline, the 23 experiment scripts, the submitted test-output file, and reports recording each measurement including the negative ones, plus three audit scripts: `count_conventions.py` recomputes every count in Table 2 from the development split, `validate_table3.py` recomputes *overall* from the official components of every submission in the evaluator log, and `check_manuscript.py` checks the numbers printed here against both. We redistribute no dataset and no test label; the scripts read the official release. The crowded-page check of §6 was run ad hoc and is described in the reports rather than packaged.

10 Limitations

The most important limitation is what our headline number represents. Our best *fully automated* run scores **0.5519** (v9 in Table 3); the reported 0.7649 adds per-question auditing against source PDFs and corrections attributed using leaderboard feedback. The audit is not a model that would run on new questions, and the attribution is adaptation to this split. A reader interested in what an automated system achieves on LitTraceQA should read the 0.5519 row. We report both prominently for that reason, and include the submission that regressed rather than only our best.

Second, the conventions in §4 are counts over 55 development questions and several rest on small denominators (3/3, 8/8, 10/10). They were predictive on test, which is evidence, but are not established at any conventional level of confidence.

Third, part of what the metric measures is *reproduction of the annotator’s bookkeeping* rather than grounding. Two mechanisms do this. Evidence keys include the visible object id, so a system locating the correct claim on the correct page of the correct paper scores **zero** if it writes Table 2 where the annotator wrote Table 3; Table 1 is that case, costing a full 1.0 of evidence F1 on one question. Table cell accuracy is gated on exact row-key matching, so a question whose invented key wording we cannot reproduce contributes zero to *two* of the five metrics however correct the extracted values are; we verified values for several

such questions and still scored zero on both. Our record of 0 for 13 on guessing those strings is evidence that this component is not reducible by better reading.

This does not invalidate the benchmark, since provenance should be scored and coarse keys are a reasonable way to do it, but the table components measure something different from the evidence components, and a system’s score can move without its grounding improving. A future edition might publish the row-key vocabulary alongside the schema, accept a small set of canonical object-id spellings, or score cells against gold rows independently of key matching. Any of the three would make the answer metrics measure answering.

Fourth, we did not submit to `littraceqa-test-extra`, so we cannot report how these findings transfer to its larger question pool.

11 Ethics

Papers were retrieved from publisher-provided URLs in the released metadata and used only for evidence extraction, within the dataset’s CC BY-NC 4.0 terms. Such systems make claims about literature checkable by pointing at a specific page of a specific paper, which we consider beneficial. The risk is that a confidently formatted citation persuades even when wrong, and our error analysis shows the dominant failure is a citation that is *almost* right: the correct paper and an adjacent row. Such systems should surface the located span for confirmation rather than presenting a grounded answer as settled.

12 Conclusion

Team OdeD’s LitTraceQA submission reached an official 0.7649 from a 0.4563 start. Two-thirds of the metric is provenance rather than answering; the annotators’ conventions are recoverable from 55 examples; and a macro metric over 71 questions is precise enough that a controlled submission can establish that an individual prediction is wrong. We also report what the process refuted, the four bugs it exposed in our verifiers, and how much of the score reflects per-question auditing rather than a system that would transfer.

Acknowledgements

We thank the GroundLM 2026 organisers for releasing the evaluator and validator.

References

- Anthropic. 2026. [Claude models overview](#). Hosted API, accessed August 2026.
- Artifex Software, Inc. 2026. [PyMuPDF: Python bindings for MuPDF](#). Software; used for page text, caption indices and page rasterisation. Accessed August 2026.
- Beijing Academy of Artificial Intelligence. 2026a. [BGE: BAAI general embedding, bge-large-en-v1.5](#). Model weights, accessed August 2026.
- Beijing Academy of Artificial Intelligence. 2026b. [BGE reranker, bge-reranker-base](#). Model weights, accessed August 2026.
- Google. 2026. [Gemini api: models](#). Hosted API; model identifiers and availability. Accessed August 2026.
- GroundLM 2026 shared task organisers. 2026. [GroundLM 2026 online evaluator](#). Hugging Face Space; source of record for all held-out scores reported here.
- LitTraceQA shared task organisers. 2026. [LitTraceQA: dataset release](#). Hugging Face dataset, CC BY-NC 4.0.
- Xuye Liu, Yimu Wang, Peng Shi, Bo Xue, Xiangrui Ke, Songcheng Cai, Kath Choi, Di Wu, Freda Shi, and Krzysztof Czarnecki. 2026. [LitTraceQA: A benchmark for multi-stage grounding and verification in scientific question answering](#). *Preprint*, arXiv:2608.07370.
- Qwen Team, Alibaba Group. 2025. [Qwen3-8B](#). Model weights, accessed August 2026.
- Yimu Wang, Yee Man Choi, Barry Zhang, Mozghan Nasr Azadani, Sean Sedwards, and Krzysztof Czarnecki. 2026a. [Where does the answer come from? benchmarking view-level visual evidence identification in multi-view MLLMs for autonomous driving](#). *Preprint*, arXiv:2606.09644.
- Yimu Wang, Xuye Liu, Yee Man Choi, Bo Xue, et al. 2026b. Findings of the first GroundLM shared tasks: Evaluating grounded language models across visual and scientific evidence. In *Proceedings of the 1st Workshop on Grounding Language Models: Learning Faithfully and Efficiently (GroundLM 2026)*.